

CSS452 Internet of Things

Homework 1: Introduction to ESP32

1. Do the Example 3 in Lecture Note 3. Take a video to show

- circuit connection,
- Arduino code,
- demonstration: adjust the brightness of the LED by rotating the potentiometer.

Name your video as “P1” and submit to the Google Classroom.

2. Do the Example 8 in Lecture Note 3. Take a video to show

- Arduino code,
- demonstration: the serial monitor shows the local time every one second.

Name your video as “P2” and submit to the Google Classroom. **Note that your ESP32 must be able to connect to your WiFi access point in order to connect to the Internet.**

3. Connect a wire to a touch pin of ESP32 (as shown in Example 4). Write the Arduino code such that the ESP32 will perform these 2 tasks:

- when ESP32 starts, it will connect to your WiFi access point and show its IP address on the serial monitor (similar to Example 6),
- thereafter, if you touch the wire, ESP32 will restart (look at Example 4 and Example 5).

Take a video to show

- circuit connection,
- Arduino code,
- demonstration of the above 2 tasks.

Name your video as “P3” and submit to the Google Classroom. The demonstration video is shown in the attached video “CSS452 - Homework 1 - Problem 3”.
